

Siwon Ryu

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EDUCATION

Northwestern University | Combined B.S./M.S. Program

Evanston, IL

Master of Science in *Computer Science*

Expected Jun 2027

Bachelor of Science in *Mechanical Engineering* | Concentration in *Robotics*

Expected Jun 2027

- GPA: **3.85/4.00**

- Relevant Courses: Mechatronics, Robotic Manipulation, Feedback Systems, Machine Learning, Robotics Lab

WORK EXPERIENCE

Apple,

Cupertino, CA

Hardware Reliability Engineering Intern

Feb 2026 - Aug 2026

- Developing a computer vision pipeline on high-speed video footage to automatically classify hardware impact types, using segmentation models like SAM 3
- Automated a 3-day manual process down to minutes by creating a data analysis app integrating data source APIs and reliability models

Milwaukee Tool,

Menomonee Falls, WI

Automation Engineering Intern

Jun 2025 - Aug 2025

- Rewired electronic controls and troubleshoot PLC/HMI program to extend test fixture runtime by 25%, reducing manual configuration time and adding new test actuator functionality
- Automated data workflows using Python, Pandas, and SQL, saving 60+ hours/week in manual labor and processing time while improving data accuracy and reliability for global engineering teams

Procter & Gamble,

Mehoopany, PA

Manufacturing Engineering Intern

Jun 2024 - Aug 2024

- Implemented an automated program using Excel and VBA macros to optimize 600,000+ square feet of warehouse storage, cutting \$30K/month in trailer detention costs and downtime losses
- Designed and helped fabricate a retrofitted core reworker, boosting efficiency by 5x while complying with site safety

RELEVANT EXPERIENCE & PROJECTS

Northwestern Formula SAE

Oct 2022 - Jun 2025

Braking/Uprights Team

- Optimized brake rotor and rear upright designs with CAD and validated safety factor through FEA
- Refined a MATLAB program to simulate braking scenarios and document component specifications for calipers and master cylinders for best vehicle performance and safety

Computer Vision Based Robot Tracker

May 2025 - Jul 2025

- Developed an autonomous tracking system using Python and ROS 2, implementing custom nodes for a 20 Hz control loop for real-time Sphero Mini tracking
- Built a computer vision system with OpenCV color detection that feeds a PID controller over ROS 2 topics and integrates a Bluetooth LE driver for communication

Robot Arm Motion Control Simulation

Nov 2024 - Dec 2024

- Generated robotic motion paths and implemented control algorithms for a mobile manipulator robot (KUKA youBot), integrating forward kinematics, odometry, and feedback control to optimize precise pick-and-place tasks
- Conducted testing of control strategies, using CoppeliaSim for dynamic simulations and applied singularity avoidance

Dynamic Mechatronics Follower Robot

Jun 2024 - Jul 2024

- Developed an autonomous follower robot using Arduino and a differential drive system with an IR beacon transceiver to track and follow a moving beacon source
- Implemented dynamic motion control with ultrasound sensor for obstacle detection and automatic stops

SKILLS

- **Software:** C/C++, Python, ROS 2, Linux, Raspberry Pi, Arduino, MATLAB, Bash, AWS, SQL, Git, Java, JavaScript
- **Other:** CAD (SolidWorks, NX), FEA, Machine Tools, CoppeliaSim, OpenCV, PLC, Ladder Logic, Simulink